Gayatri Vidya Parishad College of Engineering Autonomous

Information Technology | Course Code: IT202

Software Engineering & Design Patterns - Software Architecture & Gang of Fo

---

Overview: Creational, Structural, and Behavioral patterns Singleton, Factory, Adapter, Decorato

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Process Models & Agile Methodology
Waterfall, Spiral, V-Model, Agile Manifesto, Scrum sprints, User stories, Burndown charts, Continuou

Unit 2: Architectural Patterns & SOLID Principles
Single Responsibility, Open-Closed, Liskov Substitution, Interface Segregation, Dependency Inversion

Unit 3: Gang of Four Design Patterns
Singleton, Abstract Factory, Builder, Adapter, Facade, Decorator, Strategy, Observer, State design p

Unit 4: Quality Assurance & Testing Strategies
Unit testing, Integration testing, Black-box boundary value analysis, White-box cyclomatic complexit

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Draw a UML Class Diagram illustrating the Observer Design Pattern and discuss how it enforces loose co
2. Compute the Cyclomatic Complexity VG for a given control flow graph using edge-node, predicate, and re
3. Contrast the Strategy Pattern with the State Pattern and explain runtime behavioral polymorphic switch

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides - Design Patterns: Elements of Reusable O
* Ian Sommerville - Software Engineering Pearson 10th Ed

Verified NoteVault Institutional Archive - Authorized for Registered Student Use Only